Robot Dataset Audit Report

SRA Robotics | SAMPLE audit of a public dataset | Prepared by: SRA Robotics | founder@sra-robotics.com

Dataset: MaxT90/so101-pick-place (public on Hugging Face) **Audited:** Oct 11, 2026 **Robot:** SO-101 arm (so_follower), pick and place	**Size:** 50 episodes, 22,771 frames, 30 fps **Cameras:** 2 (wrist, overhead) **Link:** huggingface.co/datasets/MaxT90/so101-pick-place

VERDICT: PASS after 1 fix	**49 of 50 episodes usable (98%)** Meets the 85/15 rule (at least 85% usable)	**0 Critical | 1 Major | 2 Minor** Fix: trim the first ~16 s of episode 0 (or exclude it). No re-recording needed.

Legend. Episodes: **PASS** = usable | **REVIEW** = a human should look, your call | **FAIL** = fix or drop.
Findings: **Critical** = the robot could learn something dangerous or wrong | **Major** = this episode needs fixing | **Minor** = good to fix, won't break training.

What to do next

1. Trim the first ~16 s of episode 0, or exclude it.
2. Upload the calibration file with the dataset.
3. Close the gripper before stopping each recording (episode 45).
4. In future collection, widen variety: more object positions, objects and lighting.

Findings

Level	Finding	Episodes	Evidence
Critical	None	none	-
Major	**Dead start:** robot completely still for 0-16 s, then does the task normally. A policy may learn to freeze before acting. Fix: trim the first 16 s or exclude.	0	Length check FAIL (948 frames = 31.6 s vs normal 14.6 s); watched in viewer; movement 0.0 for seconds 0-15
Minor	**No calibration file:** nothing in meta/, so the joint scale can't be verified. Fix: upload the calibration file.	All	shoulder_lift, wrist_flex, wrist_roll go past ±100; calibration checklist 1/5
Minor	**Gripper left open at the end:** ends at 14.5 vs usual end 0.8. Operator did not close it (not a hardware fault); the task finished. Fix: keep it, or re-record closed.	45	End-pose hint (REVIEW); commanded action about 14.8; watched in viewer

Coverage of this review

What	How much
Automated checks	**50 of 50 episodes** (9 checks, including full video decode: 22,771 of 22,771 frames on both cameras)
Watched by a human	**2 of 50 episodes** (episodes 0 and 45) in this sample
Calibration checklist (5 yes/no)	Done by hand: 1/5
Not measured in this sample	Label review (full) and the coverage score (object position, objects, lighting)

How this was checked (reproducibility)

Tool: robot-dataset-audit (audit_checks_v1; 9 automated checks + 1 end-pose hint) | **Date run:** Oct 11, 2026
Dataset revision: see dataset revision at audit date | **Run:** `python audit_checks_v1.py MaxT90/so101-pick-place --videos`
The dataset may have changed since the audit date.

Checks run in detail

Check	What it asks	Result
Episodes present, none empty	Are all episodes present, in order, none empty?	**PASS**: all 50 present (0-49), shortest 346 frames
Duplicates	Is any episode an exact copy of another?	**PASS**: 50 unique
Lengths	Is any episode under half or over double the normal length?	**FAIL**: episode 0 is 948 frames vs normal 438 (14.6 s); limits 219-876
Timing	Steady 30 fps, no skipped frames?	**PASS**: steady 30 fps, 0 skipped
Sync	Are the cameras in sync with the robot data?	**PASS**: 0.000 s max gap
Videos	Does every video decode end to end?	**PASS**: 22,771/22,771 frames on both cameras
Action vs state	Does the robot follow its commands?	**PASS**: mean gap 0.9-3.1; gaps above 20 in at most 0.6% of rows
Actions	No dead joints; clipping under 1%?	**PASS**: no dead joints, clipping under 1%; shoulder_lift, wrist_flex, wrist_roll go past ±100
Empty episodes	Does every episode actually move?	**PASS**: least-moving episode (24) is 88% of typical
End-pose hint	Does any episode end oddly (e.g. gripper open)?	**REVIEW**: episode 45, gripper left open (task finished)

Human review

Item	How	Result
Endings	Watched the episode flagged by the end-pose hint	**REVIEW**: episode 45, gripper left open; task finished
Long episode	Watched episode 0 in the viewer; confirmed with per-second data	Dead start confirmed (**FAIL** until trimmed)
Calibration	Calibration Presence Checklist v1 (5 yes/no), by hand	1/5 (only camera details present)
Camera clear, label correct	Sampled human video review	Not fully reviewed in this sample
Coverage	Coverage Rubric v1	Not scored in this sample (part of the full audit)

Unusable episodes

Episode	Reason	Fix
0	Dead start: robot still for the first 16 s	Trim the first 16 s, or exclude the episode

49 of 50 usable (98%), passes the 85/15 rule (at least 85% usable, i.e. at most 7 bad episodes out of 50). Episode 0 is unusable until trimmed.

Limits of this audit

Automated checks cannot catch wrong labels or frozen-but-decodable camera pictures. Those are covered only by sampled human video review (2 of 50 episodes watched here: 0 and 45). Labels were not fully reviewed in this sample.

Glossary

Term	Meaning
Episode	One recorded attempt at the task (one demonstration).
Frame / fps	One time step of data; frames per second (this dataset: 30).
Dead start	The robot sits still at the beginning of an episode before the task begins.
Clipping	A joint value stuck at its limit, so real motion is cut off.
85/15 rule	At least 85% of episodes usable, at most 15% bad.
Calibration file	Data that says what the joint numbers mean, so others can reuse the dataset safely.
meta/	The dataset folder holding its description files (such as info.json).